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Article

A Multimodal Fusion Framework for NF/RO Performance: Leveraging Molecular Graphs for Superior Predictive Modeling and Mechanistic Explanations

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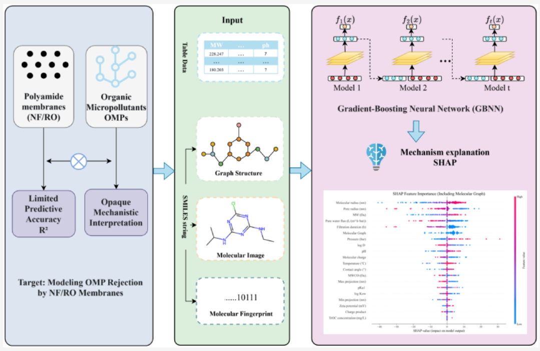

ABSTRACT: This study tackles two bottlenecks—limited predictive accuracy and opaque mechanistic interpretation—in modeling the rejection of organic micropollutants by nanofiltration/reverse-osmosis (NF/RO) polyamide membranes and introduces a multimodal-fusion machine-learning model within a gradient-boosting neural-network framework. By systematically integrating conventional tabular inputs—membrane structural parameters, operating conditions, and pollutant physicochemical properties—with multimodal molecular representations (graphs, images, and fingerprints), we developed three fusion schemes. The model that fuses tabular features with molecular-graph descriptors (GB+Graph) delivered the best performance (R^2 = 0.9014), clearly surpassing the tabular-only baseline (R^2 = 0.8494) and the other fusion variants, indicating that molecular topological information is pivotal for improving predictive accuracy. Further interpretability analysis using SHAP showed, in the global feature-importance ranking, the dominant effects of steric-hindrance-related parameters (e.g., molecular radius, molecular weight, and membrane pore size) and quantified the contributions of electrostatic interactions and hydrophobic effects to rejection. Notably, atom-level attribution analysis revealed that the model automatically highlights the central role of key functional groups in molecule–membrane interactions, providing a microstructural mechanistic interpretation beyond traditional descriptors. Overall, this work offers an advanced multimodal-fusion paradigm for accurate prediction of organic-micropollutant rejection while deepening the understanding of the coupled, multimechanistic retention process in polyamide membranes, thereby furnishing a theoretical basis and data-driven pathway for the targeted design of membranes to efficiently remove specific pollutants.

KEYWORDS: *organic micropollutants, NF/RO, multimodal fusion, machine learning*

1. INTRODUCTION

Effective removal of trace organic micropollutants (OMPs) (including pharmaceuticals and personal care products, PPCPs, and endocrine-disrupting compounds, EDCs) from aquatic environments is critical to safeguarding water resources. The urgency stems from the widespread occurrence of OMPs in freshwater systems: although typically detected at nanogram-per-liter to microgram-per-liter levels, their poor biodegradability means that chronic exposure can cause reproductive disruption in aquatic organisms, induce antibiotic resistance, and ultimately threaten human health.[1−4] For example, in some source waters, concentrations of EDCs such as estrone and pharmaceuticals such as diclofenac have exceeded potential risk thresholds, whereas conventional treatment processes often fail to remove them effectively.[5]

Nanofiltration (NF) and reverse osmosis (RO) membranes with polyamide (PA) selective layers are key technologies for advanced OMP removal owing to their subnanometer pore sizes and high selectivity, and they are widely used in water reuse and drinking-water purification. However, most commercial PA membranes were originally designed for desalination and are not optimized for the low molecular masses and complex charge properties of the OMPs. Consequently, the rejection of certain hydrophobic OMPs can be below 20%, falling short of advanced treatment needs.[2,6] Mechanistically, OMP rejection by PA membranes is governed primarily by steric exclusion (comparison of pore size and solute size), electrostatic interactions between membrane surface charge and solute charge, and affinity effects such as hydrophobic interactions and hydrogen bonding. Although several theoretical models, such as the solution–diffusion model, describe the rejection process, they typically rely on

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idealized assumptions (for example, structural homogeneity of the membrane). As a result, their generalizability across different PA membranes (for instance, dense RO versus looser NF) and diverse OMP classes is limited, which constrains the precise design and development of membranes targeting efficient OMP removal.[7]

In recent years, machine learning (ML) has shown considerable promise for predicting NF/RO rejection of OMPs, elucidating mechanisms, screening membrane materials, and optimizing fabrication processes.[8−12] To enhance model performance and physical relevance, existing ML studies in this field have adopted various strategies: some introduced mechanism-inspired features (e.g., "size ratio" for steric hindrance and "charge product" for electrostatic interactions,[13] or features related to size exclusion and charge polarization[27]), while others applied explainable AI (XAI) tools such as SHAP, permutation-based variable importance (PVI), and partial dependence plots (PDPs) to decipher key factors.[8,10,16,28] For example, Zhu et al.[10] found XGBoost to be the best-performing algorithm on a large-scale dataset and identified molecular volume as a key descriptor via SHAP, while Lu et al.[16] clarified the dominant roles of membrane pore size and zeta potential in ion selectivity using random forests and XGBoost. Nonetheless, ML applications in this domain face persistent challenges, largely due to the scarcity of high-quality experimental data, particularly for certain high-risk types of OMPs. Moreover, most existing ML studies remain confined to conventional tabular data—including aggregated membrane structural parameters (e.g., pore size, zeta potential), operating conditions (e.g., pressure, temperature), and basic physicochemical properties of OMPs (e.g., molecular weight, logKow)—and have not systematically mined the rich structural information intrinsic to OMP molecules (e.g., precise spatial conformations, functional-group distributions, and electronic properties). This limitation, coupled with the scarcity of high-quality experimental data, hinders ML models from fully capturing the intrinsic links among molecular structures, operating conditions, and rejection behavior, especially when multiple mechanisms are coupled or confounders are present. While data-driven models are capable of learning nonlinear interactions and coupled effects with sufficiently large, representative, and well-designed datasets, the practical scenario of OMP rejection prediction is often constrained by limited data volume and reliance on single-modal tabular data. Under such conditions, purely data-driven ML models often fail to reveal the underlying pathways in a complete and transparent manner.

Prior studies indicate that embedding domain knowledge—for example, by constructing inputs consistent with physicochemical laws (e.g., membrane pore size, solute charge for capturing steric exclusion, electrostatic interactions, and affinity effects) and integrating classical rejection models—into ML frameworks can mitigate data limitations and enhance physical interpretability and generalization.[13,14] Specifically, this hybrid modeling interfaces with the ML architecture via three key pathways: embedding mechanistic features in the input layer, refining mechanism-relevant information through modality-specific subnets, and validating and deciphering mechanisms via SHAP-based interpretive feedback. Therefore, integrating mechanistic insights with ML (e.g., via multimodal molecular representation) can enable accurate prediction of OMP rejection and deepen understanding of PA-membrane rejection mechanisms, offering substantial value for advancing NF/RO technologies in trace-pollutant control.[15−18]

Multimodal molecular representation learning, defined as the concurrent use of diverse data forms such as molecular graphs, sequence strings, images, and fingerprint descriptors, has achieved notable success in molecular property prediction[20−22,26] and drug discovery.[23−25] This approach has overcome the limitations of single modalities by combining complementary views, and several effective fusion pathways have been validated in related fields: for structural fusion, Chen et al.'s MolGT framework[19] fuses 2D topology with 3D geometry via a graph Transformer; for sequence-graph integration, Gong et al.'s MDFCL[20] uses contrastive learning to combine SMILES sequences with molecular-graph features, and Rollins et al.'s MolPROP[21] fuses sequence features from ChemBERTa-2 with topological features from graph networks; Nguyen et al.[23] even developed a framework that learns 3D conformations from SMILES inputs under cross-modal guidance, and ImageDDI[25] incorporates molecular images to capture spatial-distribution details. For example, fusing molecular graphs that capture topology with SMILES sequences that encode functional-group information can learn structure−activity relationships more precisely and improve activity prediction.[20,25] Combining molecular images with 3D geometric information can better characterize surface features and conformational dynamics, thereby enhancing property-prediction models.[23]

Recent advances in ML for membrane rejection prediction have begun to explore molecular structure encodings and hybrid modeling frameworks. Notably, several studies have successfully incorporated Simplified Molecular Input Line Entry System (SMILES) strings or molecular fingerprints as complementary inputs to tabular membrane and operational data, achieving improved prediction of OMP rejection by NF/RO membranes.[45,46] Furthermore, the integration of physically inspired parameters from classical models into ML frameworks—often termed physics-informed machine learning (PIML)—has enhanced the physical interpretability and generalization of predictions.[47,48] These works, frequently employing powerful algorithms like XGBoost and interpretation tools like SHAP, have significantly advanced our understanding of key factors such as electrostatic interactions, steric hindrance, and the role of specific functional groups.[45,48]

However, critical gaps remain, particularly regarding the systematic exploitation of multimodal molecular information and the depth of the mechanistic interpretation. First, while the aforementioned studies demonstrate the value of incorporating a form of molecular representation (e.g., SMILES sequences or fingerprints), a systematic comparison of different molecular modalities—specifically, molecular graphs that preserve atomic connectivity and local topology, molecular images that capture spatial geometry, and structural fingerprints—within a unified prediction framework for membrane rejection is lacking. The relative merits and complementary information provided by these distinct views of molecular structure are unexplored. Second, most existing models rely on standard gradient-boosted trees or separate neural networks for feature extraction, leaving room for architectural innovations better suited to fusing these heterogeneous data modalities. Third, although functional group importance has been identified, interpretation granularity has rarely reached the atomic level, which is crucial for pinpointing the exact structural motifs that dominate solute-membrane interactions.

To address these limitations, this study proposes a multimodal-fusion ML model grounded in a gradient-boosting framework, with the goal of systematically improving both prediction and mechanistic analysis of the NF/RO rejection of the OMPs. We first established a baseline model using a gradient-boosting algorithm. We then design and implement three fusion schemes: tabular data plus a molecular graph, tabular data plus a molecular image, and tabular data plus a molecular fingerprint. Using the tabular-only model as a control, we compare the predictive accuracy for the rejection of the OMP through cross-validation. We further apply SHapley Additive exPlanations (SHAP) to interpret the best-performing fusion model, quantifying the contributions of membrane parameters, operating conditions, physicochemical properties, and multimodal features and identifying the hierarchical key factors that govern rejection efficiency. By elucidating feature-contribution patterns, this study aims to deepen the understanding of the coupled mechanisms by which PA membranes reject OMPs and to provide a theoretical basis for targeted NF/RO membrane design and process optimization for deep removal of specific OMPs.

2. METHODOLOGY

2.1. Data Collection and Description. The dataset used in this study was constructed from experimental data on membrane removal of organic contaminants reported in two published studies.[10,27] To ensure data reliability and suitability for subsequent analyses, we first integrated the removal datasets from both sources——aligning the formats of core parameters such as membrane type, contaminant class, and rejection rate to ensure consistency—and then rigorously excluded rows with excessive missingness in key fields, including membrane performance parameters, contaminant physicochemical properties, and rejection. The final curated dataset comprises 1618 membrane-contaminant data points, covering 35 nanofiltration (NF) and reverse-osmosis (RO) membranes and 170 organic contaminants, thereby providing a standardized and complete basis for subsequent membrane-performance analysis and predictive modeling. The collected dataset can be viewed in "data.xlsx". The list of features contained in the data is presented in Table S2.

2.2. Multimodal Molecular Feature Engineering. As a standardized digital representation of molecular structure, SMILES (Simplified Molecular Input Line Entry System) can be converted via the RDKit cheminformatics toolkit[29] into three canonical forms of representation: molecular graphs, molecular images, and molecular fingerprints. These three modalities characterize molecular essence from the perspectives of topology, visual morphology, and fragment features, respectively, thereby providing diverse inputs for subsequent targeted feature extraction and property prediction.[30]

2.2.1. Molecular Image. In recent years, molecular imaging has attracted wide attention as an emerging form of molecular representation,[31,32] rendering chemical structures as pixel-level visual data that directly reflect spatial geometry and morphological features. We parse SMILES into molecular objects using RDKit and generate 224 × 224 RGB images with MolToImage to retain stereochemical information and the spatial arrangement of functional groups, thereby providing an important spatial complement for molecular property prediction.

Molecular images are highly sparse: more than 90% of pixels are background, and only small regions carry an informative structure. Accordingly, we apply center cropping to standardize images to a fixed size, preserving key structures while ensuring input consistency. Given the broad spatial distribution and relative sparsity of molecular features, we adopt a Transformer architecture[33] as the feature extractor, which globally models relationships across image regions and efficiently captures long-range dependencies between atoms. The resulting molecule-level feature vector furnishes a globally consistent and robust visual representation for downstream property prediction.

2.2.2. Molecular Fingerprint. To obtain molecular structural features, we adopted molecular fingerprints as a third representation. By encoding molecular structures into fixed-length binary or numerical vectors, fingerprints effectively capture composite molecular characteristics and provide complementary information.[34] Among various fingerprint types, we found that the MACCS fingerprint[35] performed best for the present task and therefore selected it as our fingerprint feature.

The MACCS (Molecular Access System) fingerprint is a rule-based structural-key fingerprint comprising 166 predefined molecular substructure features. Each feature corresponds to a specific chemical substructure or property—such as the presence of a particular functional group or a ring motif. When a molecule contains the corresponding substructure, the associated bit is set to 1; otherwise, it is 0, yielding a 166-dimensional binary vector. To address the high-dimensional sparsity of MACCS fingerprints, we apply a fully connected neural network (FCNN) to transform the features, producing a fixed-dimensional dense vector as the final output.

2.2.3. Molecular Graph. For a set of molecular graphs $\mathcal{G} = \{G_1, G_2, ..., G_N\}$, each single molecular graph $G_i = (V(G_i), E(G_i))$ has a node set $V(G_i)$ corresponding to the molecule's m atoms and an edge set $E(G_i)$ corresponding to the k chemical bonds between atoms. An edge $e_{uv} \in E(G_i)$ denotes the chemical connection between atoms u and v. These molecular graphs are obtained by directly converting the pollutants' SMILES strings using RDKit, which automatically assigns to every node and edge a feature vector containing chemical attributes; the details of these features are provided in Table 1. Compared with classical molecular descriptors (e.g., Balaban J, molecular weight), molecular graph representations offer unique advantages for membrane rejection modeling:[1] they preserve fine-grained structural integrity, including atomic connectivity, bond types, and local chemical environments of functional groups—details lost in aggregated scalar descriptors,

Table 1. Characteristics of Nodes and Edges of Molecular Graphs Generated by RDKit

Type	Feature	Size	Description
Node	Atomic num	119	Atomic type
	Chirality list	5	Chiral isomer
	Degree list	12	Degree of nodes
	Charge list	12	Charges on atoms
	numH list	10	Adjacent hydrogen atoms
	Radical e list	6	Radical electrons
	Hybridization list	7	Hybridization orbital type
	Is aromatic	2	Aromatic atom or not
	Is in ring	2	In ring or not (atom)
Edge	Bond type list	5	Bond type
	Bond stereo list	6	Bond stereochemistry
	Is conjugated	2	Conjugated bond or not

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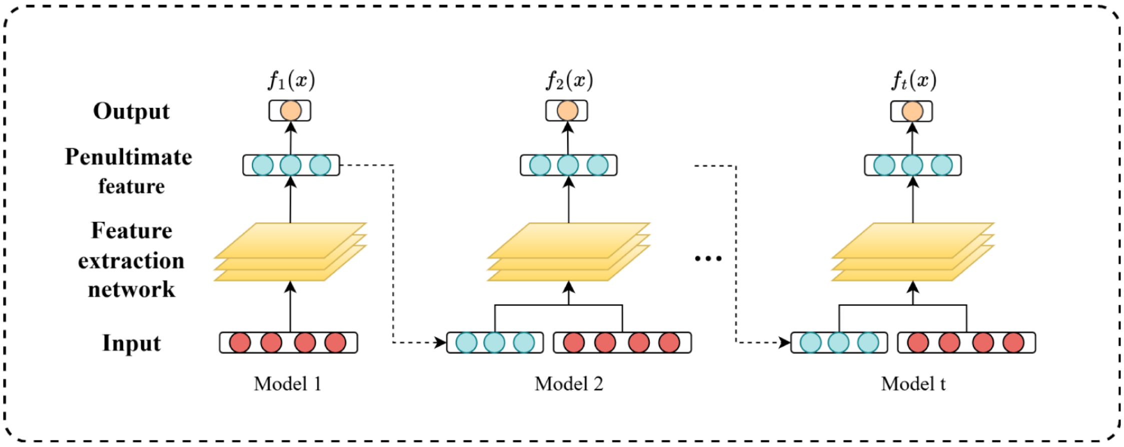

Figure 1. GrowNet: gradient boosting network architecture diagram.

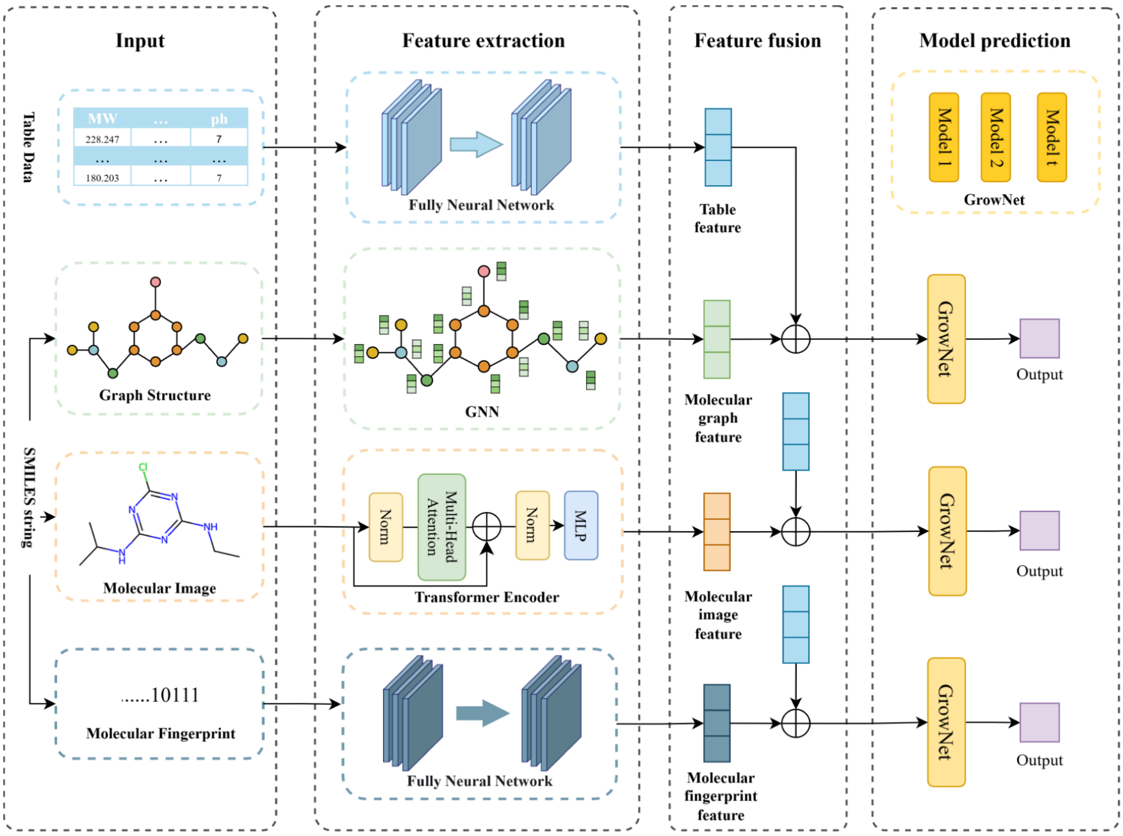

Figure 2. Architecture diagram of gradient boosting neural network and multimodal data fusion.

which are critical for distinguishing membrane-pollutant interaction differences between molecules with similar global properties.[2] Combined with GNNs and SHAP-based atom-level attribution, they enable microscopic mechanistic interpretation, pinpointing key functional groups (e.g., −COOH in ibuprofen) that dominate rejection, a capability beyond traditional descriptors limited to global summaries.[3] They inherently integrate multidimensional structural information (e.g., atomic charge, conjugation, polar/nonpolar moiety distribution), facilitating the capture of synergistic effects between steric exclusion, electrostatic interactions, and affinity mechanisms that govern rejection. These strengths make molecular graphs a more effective representation for improving the predictive accuracy and mechanistic depth.

2.3. Gradient Boosting Framework and Multimodal Fusion Strategies. Badirli et al.[36] proposed a gradient-boosting framework that uses shallow neural networks as weak learners. The method follows the core principle of gradient boosting: numerical optimization in the function space by incrementally building an ensemble of weak learners. Starting from a simple learner, each subsequent learner is trained sequentially to fit the negative gradient of the current residuals, thereby reducing the overall loss and yielding a strong model. The goal is to overcome the limitations of traditional gradient-boosted decision trees (GBDT; e.g., XGBoost, LightGBM), which rely on decision trees as weak learners, and to leverage the advantages of deep neural networks for structured data.

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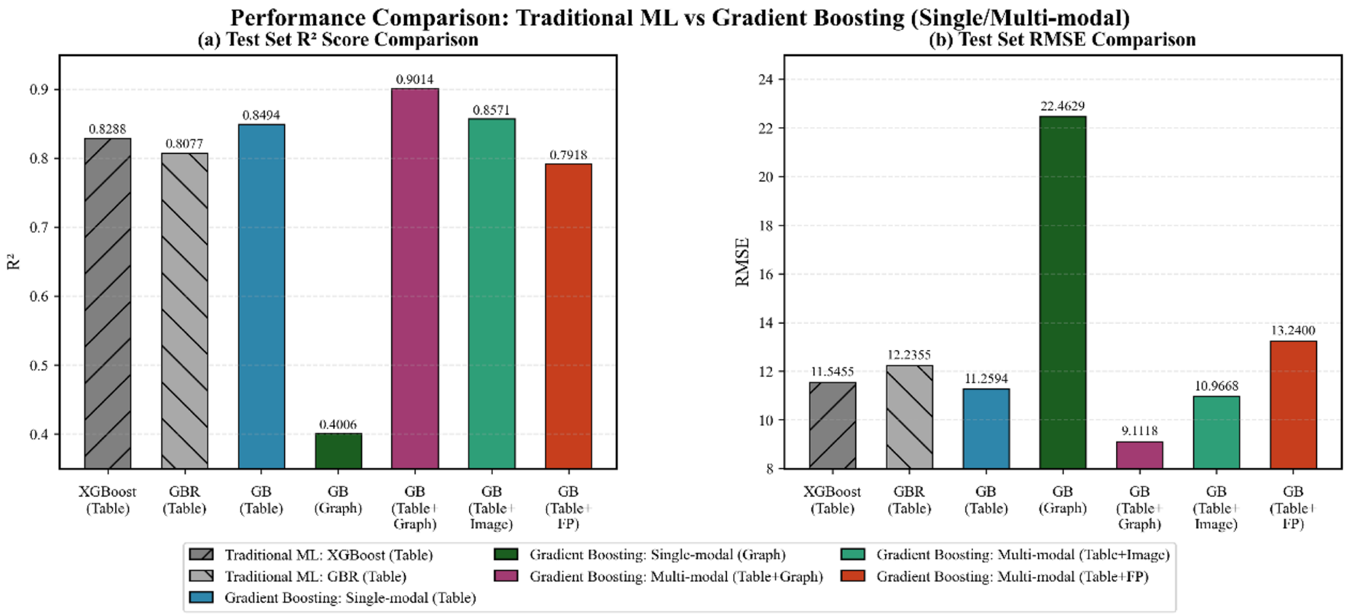

Figure 3. Comparison of R-squared and RMSE evaluation metrics for different models on the test set.

Building on this idea, our study adopts the GrowNet framework logic, replacing the decision trees in conventional GBDT with shallow neural networks as weak learners. Through feature concatenation, for example, by combining the original inputs with the penultimate-layer features from the previous weak learner, the model enhances information flow and better captures complex nonlinear relationships (see Figure 1). This design is tailored to the prediction of OMP rejection.

Multimodal fusion uses a unified early fusion strategy. For the three molecular modalities—molecular graphs, molecular images, and molecular fingerprints—we extract features with dedicated subnetworks. Graph features are produced by a graph neural network (GNN) via node encoding, message passing, and global pooling. Image features are extracted with a transformer architecture and then reduced in dimension by a fully connected neural network (FCNN). Fingerprint features (e.g., MACCS) are transformed into dense vectors by an FCNN. Each modality-specific feature vector is concatenated with the tabular features, which include membrane structural parameters, OMP physicochemical properties, and operating conditions. The fused representation is then fed into the same gradient-boosting model for training and prediction. The overall architecture is shown in Figure 2.

To quantify the incremental value of multimodal features, we set a tabular-only model as the control. Only the 22-dimensional tabular features are provided to the gradient-boosting model, while the loss function, hyperparameter ranges, and cross-validation protocol are kept identical to those of the fusion schemes. By comparing prediction error (RMSE, MAE) and the coefficient of determination (R^2), we evaluate the contribution of multimodal molecular features to OMP-rejection prediction performance.

2.4. Model Evaluation and Interpretation Protocol.

2.4.1. Performance Metrics. This study undertakes a regression prediction task. In evaluating model performance for regression, commonly used metrics include mean squared error (MSE), mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2), each with its own advantages. Specifically, MSE and RMSE are more sensitive to outliers, yet their strength lies in accurately reflecting the magnitude of deviation between predictions and ground truth; MAE exhibits robustness to outliers and provides a more stable reference when assessing model fit; and R^2, as an important statistic for assessing the goodness of fit of regression models, primarily quantifies the degree of agreement between the model and the observed data. Its values lie between 0 and 1, with higher values indicating a better fit to the observations.

To more precisely assess model performance, after due consideration, we ultimately designate RMSE and R^2 as the core evaluation metrics.[37,38] For the models constructed here, we expect higher R^2 and lower RMSE. The specific formulas are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2}{\sum_{i=1}^{n} (\overline{y}_i - y_i)^2} \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2} \tag{2}$$

where $\hat{y}_i$ represents the model predicted value, y_i represents the true value, $\overline{y}_i$ represents the sample mean, and n is the total number of samples.

2.4.2. SHAP Analysis for Mechanistic Insight. To probe the model's decision mechanism and clarify how data features influence the target variable, we adopt the Shapley value from cooperative game theory as the theoretical basis and use SHapley Additive exPlanations (SHAP) for model interpretation.[39] SHAP provides a principled way to quantify the contribution of each feature to an individual prediction and to the model's behavior overall. By attributing the prediction to features in an additive manner that satisfies efficiency, symmetry, and consistency, SHAP enables the comprehensive discovery of latent associations among features and a clear delineation of their specific effects on the target. In essence, the method decomposes a model output into a sum of feature-level

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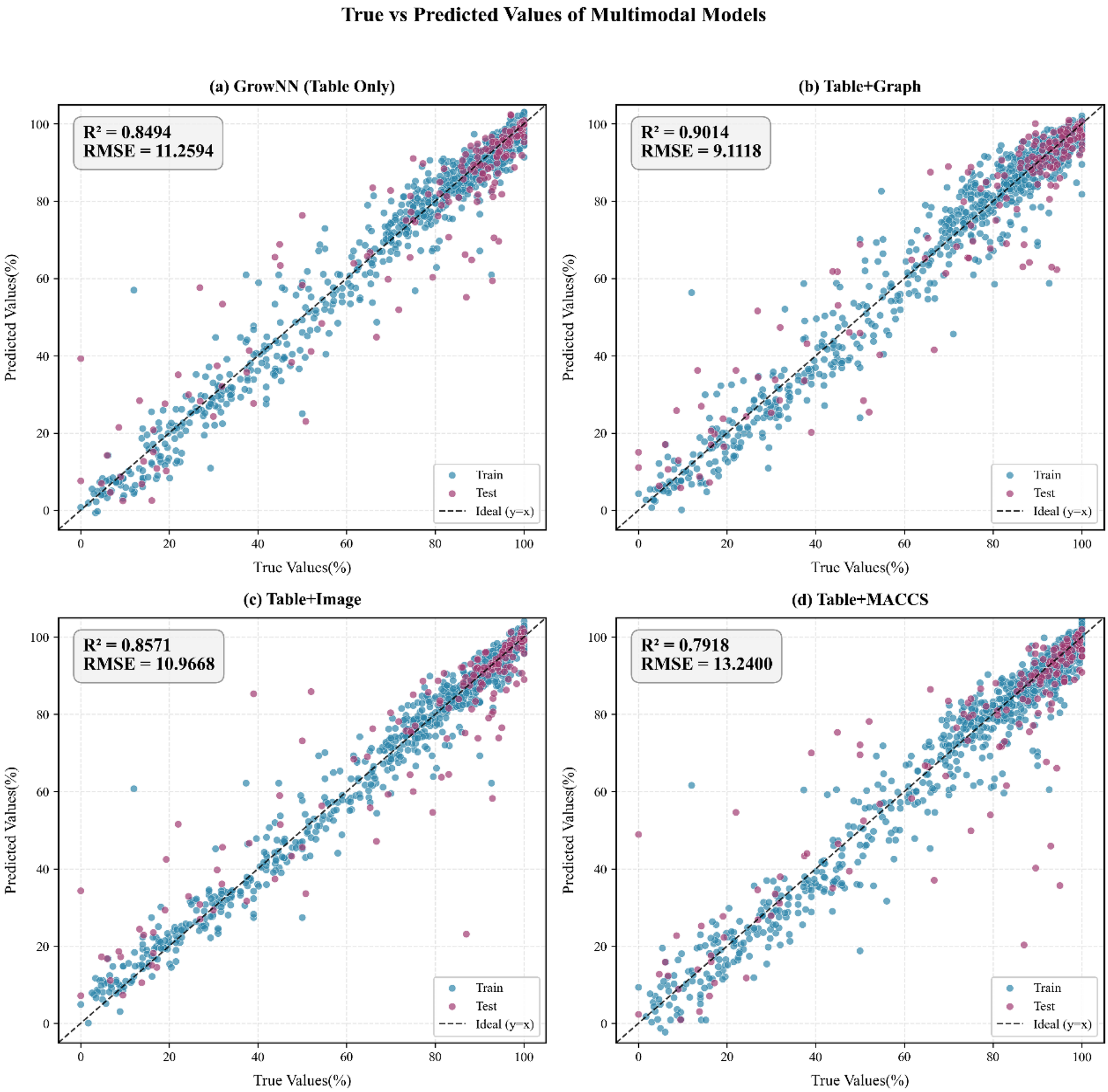

Figure 4. Scatter distribution of predicted values and true values of single-modal and multimodal models on the training and testing sets.

contributions, yielding a fine-grained local explanation of the prediction logic.

3. RESULTS AND DISCUSSION

3.1. Predictive Performance of Multimodal Models.

To systematically assess the predictive capacity of multimodal fusion strategies for polyamide-membrane rejection of OMPs, we implemented a comprehensive multitier model comparison—supplemented with a critical baseline to address reviewer comments. First, we adopted two widely used gradient-boosted decision tree baselines, XGBoost and gradient boosting regression (GBR), both trained solely on tabular data. Building on this, we introduced a gradient boosting model (GB) that uses shallow neural networks as weak learners as the core framework and evaluated its tabular-only variant. To isolate the benefit of fusing tabular data with molecular graphs (rather than relying on graph-based models alone), we further supplemented a standalone GNN model as an additional baseline: this model exclusively uses molecular graph features with the same GNN subnetwork (node/edge feature encodings, message-passing mechanisms, and global pooling strategies) as the graph branch in the GB+Graph fusion model, ensuring fairness by eliminating structural discrepancies. Finally, we tested three multimodal fusion schemes (tabular plus molecular graph, tabular plus molecular image, and tabular plus molecular fingerprint). Performance on a common test set, measured by R^2 and RMSE, is summarized in Table S3 and visualized with bar charts in Figure 3.

Even under identical tabular-only inputs, our gradient boosting neural framework (R^2 = 0.8494, RMSE = 11.2594)

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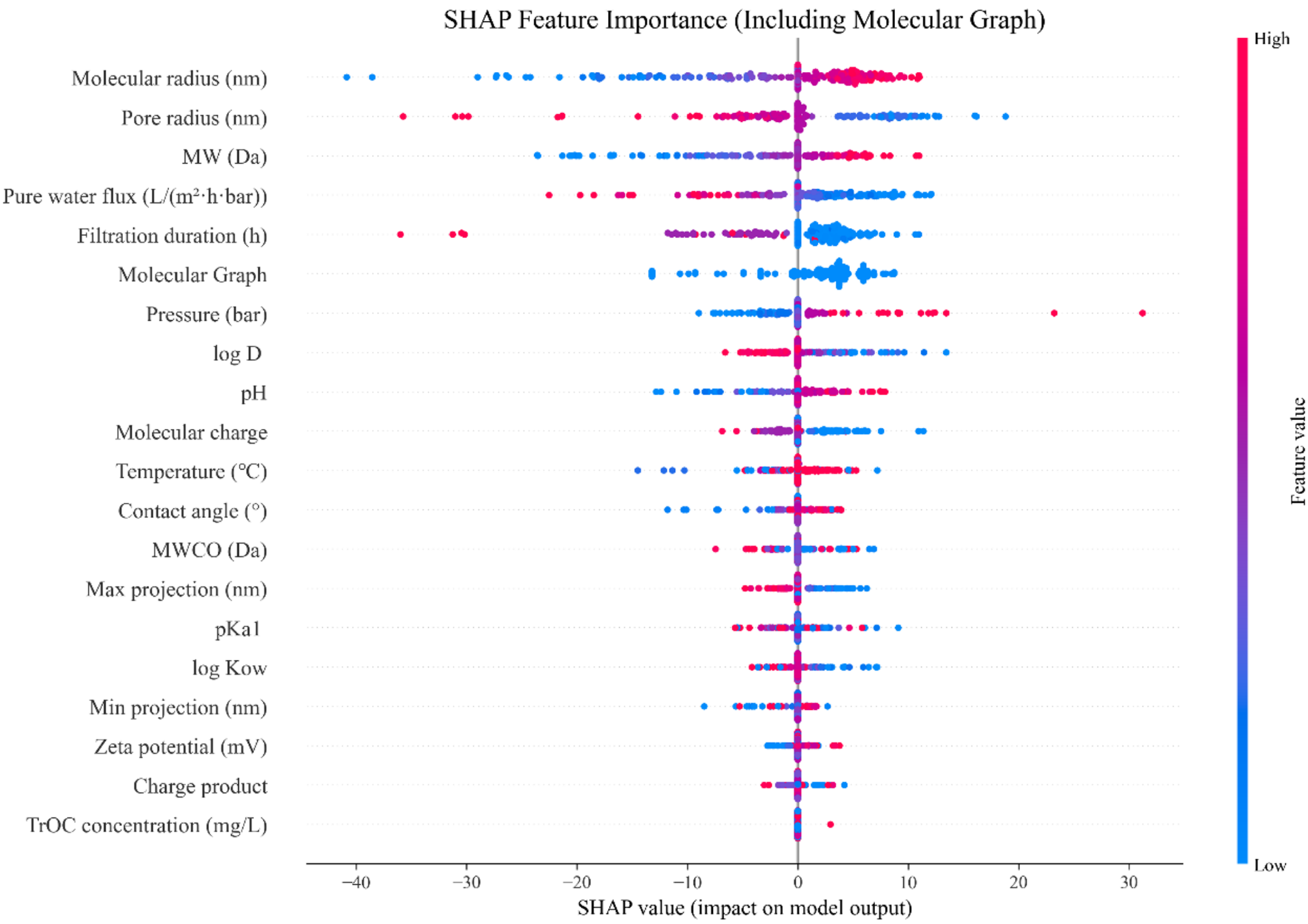

Figure 5. Global feature importance analysis based on SHAP.

outperforms XGBoost (R^2 = 0.8288, RMSE = 11.5455) and GBR (R^2 = 0.8077, RMSE = 12.2355). This result indicates that through neural weak learners and feature concatenation, the framework more effectively captures complex nonlinear relationships and provides a stronger foundation for subsequent multimodal fusion.

Notably, the standalone GNN model (exclusively using molecular graph features) exhibits the poorest performance among all models, with R^2 = 0.4006 and RMSE = 22.4629 (Figure 3 and Table S3). Its R^2 is only ∼44% of the tabular-only GB baseline and ∼44% of the GB+Graph fusion model, while its RMSE is more than twice that of the fusion model. This stark performance gap confirms that molecular graph features alone are insufficient to capture the coupled mechanisms of OMP rejection, as they lack access to critical nonmolecular information (e.g., membrane pore size, operating pressure, filtration duration) encoded in tabular data.

On this basis, performance further diverges across the multimodal models. The tabular plus graph model (GB +Graph) performs best, with R^2 increasing to 0.9014 and RMSE decreasing to 9.1118. Relative to our tabular GB baseline, R^2 rises by approximately 6.12%, and RMSE falls by 19.07%, clearly surpassing the other fusion schemes and indicating that molecular graph features supply key structural information.

The superior performance of the graph-fusion model likely arises from its efficient encoding of molecular topology and its synergistic integration with tabular domain knowledge—addressing the limitations of single-modal models. Through a graph neural network (GNN), molecular graphs capture atomic connectivity, bond types, and local neighborhoods (e.g., functional group distribution) that reflect microscale molecule-membrane interactions (e.g., hydrogen bonding), while tabular features provide macroscale mechanistic cues (e.g., membrane pore size for steric exclusion and pressure for mass transfer). This complementarity is highlighted by the poor performance of the standalone GNN model: without tabular data, molecular graphs alone cannot capture the multimechanism coupling (steric exclusion, electrostatic interactions, and affinity effects) that governs OMP rejection. By contrast, the GB+Graph model integrates both microscale and macroscale information, thereby achieving the highest predictive accuracy.

By contrast, the image fusion model (GB+Image) offers only slight gains over the tabular-only GB baseline (R^2 = 0.8571). This marginal improvement can be attributed to two interrelated factors. First, while 2D molecular renderings preserve stereochemical layout, their fixed resolution (224 × 224 pixels) and inherent sparsity may inadequately capture the subnanometer steric and electronic details critical for differentiating membrane−solute interactions. Second, the global visual features extracted by the Transformer encoder likely exhibit significant overlap with information already encapsulated in tabular descriptors of molecular size and shape (e.g., molecular radius, weight), thereby providing limited *complementary* information beyond the baseline.

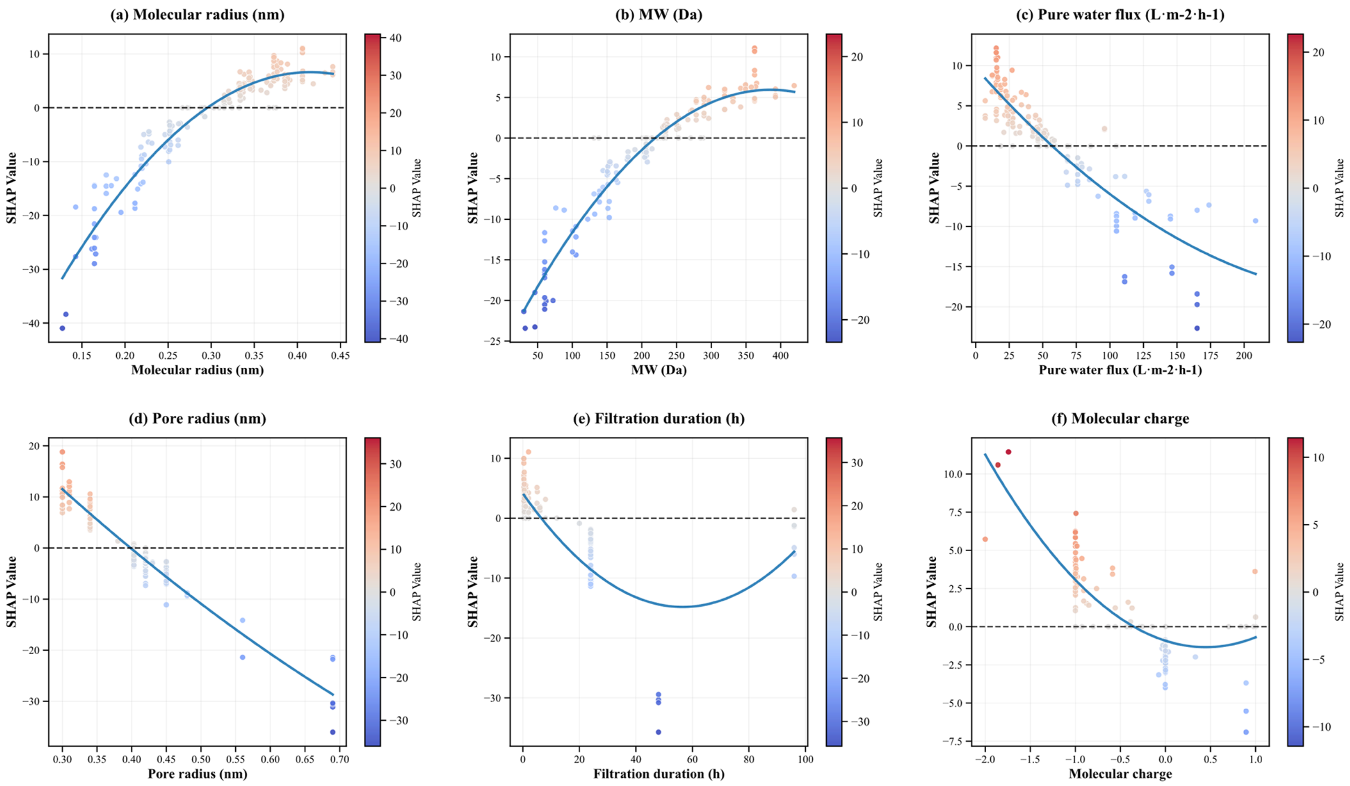

Figure 6. SHAP-based input table feature dependency map: (a) molecular radius (nm), (b) molecular weight MW (Da), (c) pure water flux (L/(m^2·h·bar)), (d) pore size (nm), (e) filtration time (h), and (f) molecular charge.

The fingerprint fusion model (GB+Fingerprint) performed worse than the baseline (R^2 = 0.7918), an outcome that necessitates a deeper analysis beyond initial assumptions. First, inherent limitations of MACCS fingerprint encoding exacerbate its underperformance. As a binary structural-key fingerprint, MACCS records only the presence or absence of predefined substructures, discarding all continuous spatial and quantitative information (e.g., the number of carboxyl groups or their three-dimensional proximity). For membrane rejection, such details are critical: two adjacent carboxyl groups induce a stronger localized negative charge and different hydrogen-bonding topology than a single isolated one, affecting electrostatic and affinity interactions in ways that the aggregate tabular descriptor "molecular charge" cannot resolve. Second, the 166-bit MACCS key set is optimized for common pharmacophores in drug discovery and may not comprehensively cover environmentally relevant OMP substructures (e.g., specific steroid rings or antibiotic side chains), leading to an incomplete molecular representation. Finally, and crucially, the fingerprint modality lacks complementarity with a major portion of the tabular dataset: it provides no information related to membrane properties (e.g., pore radius) or operating conditions (e.g., pressure), which are vital for predicting rejection. Its information is confined to the molecular structure—a domain already partially described by other physicochemical tabular features (e.g., log D, hydrogen bond counts). This results in redundancy with existing molecular descriptors without adding the complementary nonmolecular or fine-grained topological insights offered by the molecular graph. Consequently, the GB+Fingerprint fusion fails to leverage multimodal synergy and may even introduce noise, leading to a degraded performance.

To further verify reliability, Figure 4 plots predicted versus observed values for all fusion models on the training and test sets. The GB+Graph points cluster most tightly along the diagonal with high consistency between training and test results, indicating strong generalization and no apparent overfitting. Other fusion models show varying degrees of deviation, with GB+Fingerprint exhibiting more outliers on the test set, which corroborates the limitations of fingerprints for this task.

In summary, we first establish the inherent advantage of the gradient boosting neural framework for membrane-rejection prediction. Building on this, our multimodal experiments demonstrate that combining molecular graphs with tabular data most effectively improves the OMP rejection prediction. This finding confirms the pivotal role of molecular topology in membrane separation mechanisms and provides a reliable basis for a mechanistic analysis using the best-performing model.

3.2. Model Interpretation. To elucidate the decision mechanism of the optimal model (GB+Graph) and to verify the consistency of its predictions with physicochemical principles, we conducted a systematic interpretability analysis using the SHAP method. This section first examines how the model quantifies classical membrane-rejection mechanisms via global feature-importance ranking and analysis of key feature effects; it then focuses on molecular-graph features to dissect at the atomic scale the contribution pathways that govern rejection behavior.

3.2.1. Global Feature Importance Analysis. The SHAP-based global feature-importance plot (Figure 5) clearly quantifies the relative contributions of each feature to predicting the OMP rejection. The top five features, in order, are molecular radius, pore radius, molecular weight (MW), pure water flux, filtration duration, and molecular

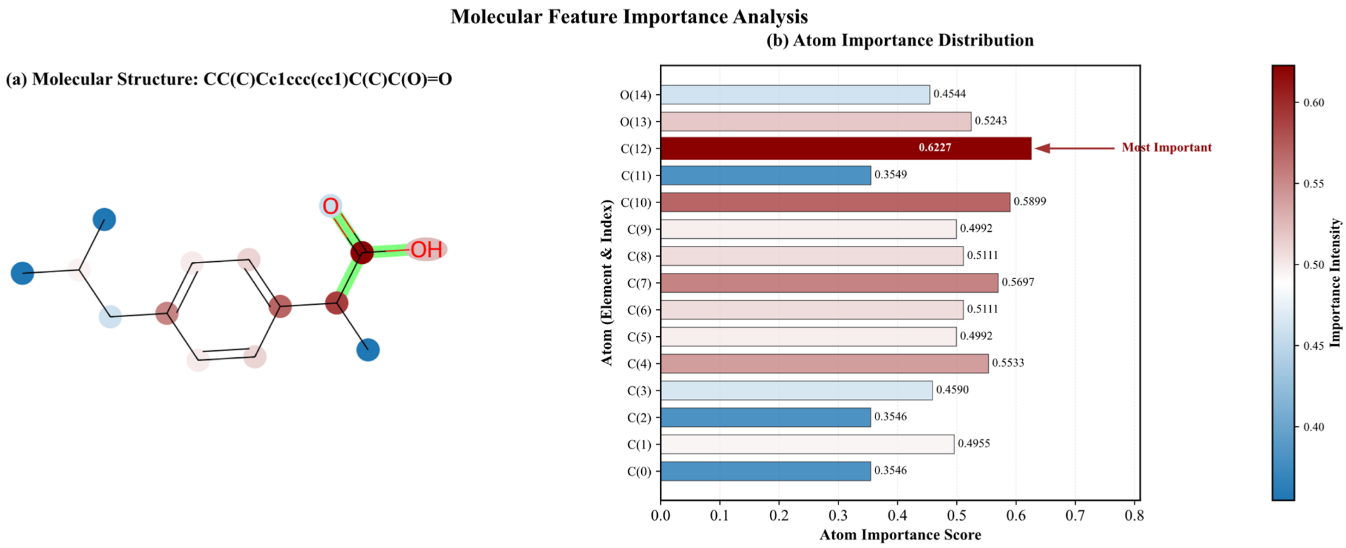

Figure 7. Using ibuprofen as an example, analysis of atomic and structural importance in the molecular diagram.

graph. Importantly, the multimodal molecular-graph embedding—a unified representation learned by the graph neural network from the atomic and bond features detailed in Table 1—ranks as the sixth most important feature overall. This ranking aligns closely with fundamental understanding in membrane separation, indicating that steric exclusion—jointly determined by solute size and pore size—is the dominant physical mechanism governing polyamide-membrane rejection of neutral and weakly charged OMPs. The substantial contribution of molecular-graph embedding directly demonstrates that the fine-grained topological information it encodes is indispensable for enhancing predictive performance. Notably, operating parameters (e.g., pressure) and solute physicochemical properties (e.g., molecular charge, log D) also rank highly, revealing that the model successfully captures the multimechanism coupling between operating conditions and molecular attributes. While the graph embedding as a whole provides a high-level summary of molecular topology, its internal importance is further dissected through atom-level attribution analysis in Section 3.2.3, which identifies key functional groups governing rejection behavior.

The dominance of steric-hindrance parameters (molecular radius, pore radius, MW) in our SHAP analysis aligns with prior ML studies on NF/RO rejection,[10,16,27] reaffirming size exclusion as a primary mechanism. A key distinction, however, is the prominent importance of the "Molecular Graph" embedding (ranked sixth overall, Figure 5). Unlike prior works that relied solely on predefined scalar descriptors, our model integrates this learned topological representation. Its high ranking demonstrates that molecular graph features capture essential structural information—complementing and extending beyond conventional physicochemical properties—for predicting rejection. This underscores the unique value of our multimodal (graph-based) fusion approach. Furthermore, our model's higher weighting of log D over log Kow for characterizing hydrophobicity[27,43] aligns with a more nuanced understanding of solute-membrane affinity interactions.

3.2.2. Effect Analysis of Key Tabular Features. By plotting SHAP dependence plots, we further dissect how individual features influence the model outputs.

3.2.2.1. Steric (Size-Exclusion) Effects. As the two most important features, the SHAP values for molecular radius and molecular weight increase with feature magnitude and thus show a strong positive association with the predicted rejection of the OMPs (Figure 6A,B). In other words, larger molecules yield higher model-predicted rejection, which reflects the core principle of steric exclusion. The effect of the pore radius is nonlinear (Figure 6D). At smaller radii, the SHAP value decreases rapidly as pore size increases because steric exclusion weakens; beyond a threshold, the decline levels off, suggesting that at larger pores, other mass-transfer processes (for example, surface adsorption−diffusion) contribute more to the apparent rejection.

3.2.2.2. Membrane Performance and Operating Conditions. Pure-water flux is positively correlated with the SHAP value (Figure 6C). Higher flux is typically associated with a denser selective layer or higher operating pressure, both of which favor enhanced rejection. Pressure shows a negative correlation, plausibly because the dataset spans a wide range of membrane types, from loose NF to dense RO; higher pressure may induce more pronounced compaction in loose NF membranes or exacerbate concentration polarization, producing a modest negative impact on apparent rejection. Filtration duration exhibits a clear inflection (Figure 6E). During the initial period (approximately 6 h), SHAP values are positive, possibly reflecting transient solute exclusion caused by initial hydrophobic adsorption within pores; after roughly 6 h, SHAP values become negative, indicating that accumulation of OMPs within pores or formation of a surface fouling layer begins to dominate,[40,41] thereby reducing effective rejection.

3.2.2.3. Electrostatic and Hydrophobic Interactions. The overall effect of molecular charge on rejection is positive (Figure 6F), consistent with electrostatic repulsion between charged OMPs and the typically negatively charged polyamide surface, which enhances rejection and is particularly important for ionic OMPs. The distribution coefficient (log D) shows a clear negative correlation: greater hydrophobicity (higher log D) corresponds to lower model-predicted rejection, in line with the conventional view that hydrophobicity can hinder TrOC removal.[42] pH affects rejection indirectly by modulating OMP speciation (and thus charge) and hydrophobicity (and thus log D);[43,44] in the model, this pathway is operationalized through these two features.

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3.2.3. Molecular Graph Feature Analysis. To further validate which structural aspects of the molecular graph contribute most to predictive performance, we conducted a systematic ablation study within the best-performing GB +Graph model. Specifically, we evaluated the impact of individual node and edge features by sequentially masking each feature type (setting its values to zero) while keeping all others intact and then retraining the model. The resulting changes in RMSE and R^2 are summarized in Figures S7−S9.

The removal of any single feature led to a decline in model performance, confirming that all encoded structural attributes provide useful, nonredundant information for prediction. However, the degree of degradation varied significantly across features, allowing us to identify the most informative ones. Among edge features, the Is conjugated attribute was the most critical (ΔRMSE = +1.71, ΔR^2 = −0.041), followed by Bond type. This highlights that capturing π-conjugation and bond order—factors influencing molecular rigidity, polarity, and potential $\pi-\pi$ interactions with the aromatic polyamide membrane—is essential. Among node features, the Charge list (partial atomic charges) had the strongest effect (ΔRMSE = +2.03, ΔR^2 = −0.049), with Degree list (atom connectivity) being the next most important. This underscores the paramount role of electrostatic interactions (governed by charge distribution) and steric configuration (influenced by atom connectivity) in determining rejection.

While the ablation study reveals which categories of graph features are most important, a deeper, atom-level understanding of how specific molecular substructures influence rejection is needed to fully interpret the model's decisions. To this end and to move beyond the limitations of conventional descriptors, we performed atom-level attribution on the molecular-graph features. This method quantifies the contribution of each atomic node in the graph to the final prediction. Quantitative methods are described in Supporting Information Text S3. As a case study, we analyzed the pharmaceutical contaminant ibuprofen. As shown in Figure 7, the analysis of atomic and structural importance indicates that the highest-contributing node is the carbon atom within the carboxyl (−COOH) functional group (highlighted in green), far exceeding other carbon or hydrogen atoms in the molecule. This result provides a valuable microscopic perspective: the model automatically identifies the carboxyl group as a key functional group that plays a central role in predicting ibuprofen's rejection.

The oxygen atoms of the carboxyl group act as prominent hydrogen-bonding sites, with the carbonyl oxygen serving as a strong hydrogen-bond acceptor; these interactions with hydrogen-bond donors on the polyamide surface may affect solute diffusion kinetics within the membrane phase. In parallel, the carboxyl group strongly influences ibuprofen's ionization state and charge under typical water-treatment pH conditions. This case shows that incorporating molecular-graph features enables the model to connect macroscopic rejection performance with microscopic molecular structure, especially critical functional groups, and provides an explanatory pathway with a higher structural resolution than traditional tabular descriptors.

To validate the generality of this atom-level interpretation, we extended the analysis to two additional OMPs with diverse properties: anionic 2-naphthalenesulfonic acid and neutral gemfibrozil (see Supplementary Figures S10 and S11 for full details). For 2-naphthalenesulfonic acid, the sulfur atom in the charged sulfonic acid (−SO_3H) group was identified as critical, aligning with the dominant role of electrostatic repulsion for this compound. Concurrently, carbon atoms at the junction of the naphthalene rings were also highly important, reflecting the contribution of the hydrophobic core to the steric and affinity interactions. For gemfibrozil, the model again highlighted carbon atoms in the carboxylic acid (−COOH) group, similar to ibuprofen, and also assigned high importance to atoms in the aromatic rings, emphasizing the role of the hydrophobic molecular backbone.

It should be emphasized, however, that the present analysis focuses on three representative molecules; establishing broader generality requires systematic atom-level studies across a more extensive set of pollutants for further validation and refinement. Nonetheless, these consistent results across molecules with different charges and functional groups demonstrate that our model's atom-level attributions can reliably identify the key structural features (charged groups, hydrophobic moieties, and aromatic systems) that govern solute-membrane interactions. This capability provides a powerful, generalizable tool for mechanistic insight on the microscopic scale.

4. CONCLUSION

This study develops a gradient-boosting-based multimodal fusion framework that integrates molecular graph information with domain knowledge from membrane separation to achieve high-accuracy prediction of polyamide-membrane rejection of OMPs. Introducing the molecular-graph modality increases predictive accuracy (R^2) by roughly 6%, outperforming models that use only tabular data and those that incorporate molecular images or fingerprints, which highlights the unique advantage of molecular topology for characterizing membrane−pollutant interactions.

SHAP-based interpretability analyses validate the dominant roles of classical separation mechanisms—steric exclusion, electrostatic interactions, and hydrophobic effects—and connect macroscale properties with microscale structure through atom-level attribution, exemplified by the key role of the carboxyl (−COOH) group in ibuprofen. The work is limited by dataset size, the absence of three-dimensional conformational information, and limited consideration of complex real-water matrices. Future research should incorporate 3D molecular conformations, develop cross-modal fusion architectures, expand datasets to complex matrices, and combine molecular dynamics simulations to test the causal validity of SHAP findings. Overall, this study demonstrates the value of multimodal fusion in membrane separation and lays a foundation for the precise design of high-performance rejection membranes and for the broader application of intelligent methods in environmental engineering.

■ ASSOCIATED CONTENT

Data Availability Statement

The data and experimental codes used in this paper are accessible through the following link: https://github.com/wang-lab-repository/MolGBN-OPR.git.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestengg.5c00938.

Detailed data processing workflow and feature representation methods (Text S1); complete architecture and training details of the GrowNet framework (Text S2);

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methodology for atomic-level attribution analysis (Text S3); list of abbreviations (Table S1); description of all tabular features used in the study (Table S2); performance metrics of all compared models on the test set (Table S3); data distribution plots for key input features (Figures S1−S5); plot of average contaminant rejection rate versus filtration duration (Figure S6); results from the ablation study on molecular graph features, showing changes in RMSE and R^2 (Figures S7−S9); and atom-level SHAP attribution visualizations for two example compounds (Figures S10 and S11) (PDF)

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Author Contributions

Y.W. conceived the idea. J.X. designed and finished the experiments, analyzed the data, and discussed the results. J.X., A.M., Y.W., and Y.W. wrote the manuscript. All authors have approved the final version of the manuscript. CRediT: **Junhua Xiao** data curation, resources, software, writing - original draft; **Ao Ma** data curation, methodology, writing - original draft; **Yunkun Wang** conceptualization, writing - original draft; **Yunqian Wang** conceptualization, writing - original draft.

Notes

The authors declare no competing financial interest.

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